

Quick Reference Guide

Module 4: Designing Artificial Machines to Solve Problems

Learning Outcomes:

- Identify the superhuman intelligence used in an AI product or process.
- Appraise the difficulties an AI technology can face.
- Identify the key features and purpose of residual networks.
- Describe how to augment a dataset using biomarkers.
- Discuss the approvals necessary for human experimentation.
- Discuss the implications of crowdsourcing data strategies.
- Demonstrate how AI can solve healthcare problems.

The Four Stages of AI Design Process for Creating Product using AI Technologies

Stage I: Figure–Ground and AI Categorization

Objective: Design an AI product that analyzes voice to assess the risk of medical conditions like Alzheimer's, Dementia, or COVID.

Identifying Superhuman Intelligence

- Visual and audio recognition
 - Visual: AI recognizing a cabin hidden by vegetation.
 - Ambiguous cases: Interpreting complex silhouettes.
 - Real-life audio segregation
 - Vehicle sounds: Detecting issues like open window noise or engine problems.
 - Mechanics' diagnosis: Identifying specific noises, e.g., valve train issues.
- AI training for sound differentiation
 - Example: Differentiating engine sounds with clean vs. dirty air filters using Fourier transform.
 - Result: Accurate diagnosis with fewer samples.

- Phone-based diagnostics
 - Microphone: Detecting air filter condition.
 - Accelerometer: Detecting wheel imbalance.
 - Future: "Shazam" for vehicle diagnostics.
- AI in medical diagnostics historical context
 - Stethoscope: Used for diagnosing conditions like tuberculosis and heart diseases.
 - Modern uses: Voice changes for detecting vocal fold polyps and other conditions.
- Challenges and opportunities
 - Cost and expertise: Traditional diagnostic tools require qualified personnel.
 - AI potential: More accessible and affordable diagnoses.
- Detectable conditions via voice:
 - Examples: Chronic Otitis, Multiple Sclerosis, Parkinson's, Hypothyroidism.
 - AI's role: Voice analysis to detect these conditions.
- Mental health diagnosis:
 - Example: Predicting psychosis onset in at-risk youths through speech analysis.
- AI in Alzheimer's diagnosis for prevalence and diagnosis
 - Statistics: Alzheimer's is common in those over 85, accounting for 75 percent of dementias.
 - Testing: Using speech tasks (e.g., describing images) to identify cognitive impairments.
- AI techniques
 - NLP: Identifying cognitive decay indicators from speech.
 - Accuracy goal: Over 99% accuracy in diagnosing and tracking Alzheimer's.
- Early detection focus
 - Importance: Early interventions are most effective.
 - Hypotheses: Amyloid plaques and Tau tangles as early indicators.

Stage II: Strategy and Operations — Alzheimer's and COVID

Alzheimer's disease: The strategy is to develop the best AI audio diagnostic tool with a target of 90% biomarker accuracy. This approach capitalizes on zero variable costs to dominate the market by providing superior diagnostic capabilities. The focus is on addressing a significant \$1 trillion global societal cost, making it a critical and impactful solution. Alternative strategies like full

customer solution and network externalities offer niche benefits and data advantages, but face limitations in scalability and legal constraints.

COVID-19: The strategy is to leverage network externalities to build the largest voice sample database for rapid and accurate diagnostics. This strategy aims to capitalize on early market entry, securing legal approvals and attracting substantial investment to quickly scale. The urgent global health need for effective COVID-19 diagnostics makes this approach highly impactful. Operationally, we will focus on rapid data collection, model development, clinical testing, and wide deployment while ensuring regulatory compliance and continuous performance optimization.

Operationally, integrating AI biomarkers into Alzheimer's trials requires engaging stakeholders like the FDA, mobilizing trials with AI assessments, investing in project management, educating doctors, and addressing support structures. This ensures effective AI integration, improving disease tracking and drug efficacy evaluation.

IP Approach: Choice of ResNets

Objective: Use ResNets for Alzheimer's AI diagnostics due to their superior architecture. Image Recognition and Processing: CNNs for facial recognition and medical imaging.

Key points:

- Model flexibility: Start with ResNet 50, switch to VGG or Densenet using PyTorch or TensorFlow.
- Architecture advantage: Forward arrow solves the vanishing gradient problem.
- Proven performance: ResNets outperform deeper traditional networks.

The Key Features and Purpose of Residual Networks

Challenges of deep neural networks

- Vanishing/exploding gradients: Gradients can shrink or grow too much.
- Degradation: More layers can reduce performance.
- Optimization: Deep networks are harder to optimize.

Understanding Residual Networks (ResNets)

ResNets solve deep network issues that was introduced by Kaiming He in 2015.

Key concepts

- Degradation solution: Residual connections improve gradient flow.
- Residual learning: Learn the difference between output and input.

- Skip connections: Bypass layers to prevent vanishing gradients.
- Building blocks: Use layers with shortcuts (Residual Blocks).
- Variants: ResNet18, ResNet34, ResNet50, ResNet101, ResNet152.

Impact

- Accuracy: Better performance on benchmarks.
- Training: Easier training of deep networks.

Applications

- Image classification: Handles complex patterns.
- Object detection: Used in detection frameworks.
- Image segmentation: Ideal for detailed tasks.

Stage III — (Part 1: IP) The Choice of the Architecture

To improve the Alzheimer's diagnostic model, we started with ResNet 50 at 60% accuracy and upgraded it by using a graph with multiple ResNets in the Open-Voice Brain Model Architecture (OVBM), which targets key brain biomarkers — Sensory Stream, Brain OS, Cognitive Core, and Semantic Models. We also applied data augmentation and transfer learning.

For COVID-19, we used a similar architecture but omitted the Cognitive Core biomarkers. The OVBM architecture enhances AI model explainability compared to ResNet 50.

Augmentation of Dataset Using Biomarker

To augment your dataset

- Use emotions and Sounds: Leverage vocal cord biomarkers to distinguish coughs and sentiment analysis to detect emotional distress, enhancing data diversity.
- Define metrics: Track vocal cord distinctiveness, sentiment impact, and cough variability to measure and improve dataset quality.

Data Strategy and IRB Approval

- Privacy and data collection: Privacy is crucial in digital health, with regulations like GDPR in Europe.
 - For Alzheimer's, we used a sanitized public dataset.
 - For COVID-19, we collected cough recordings (no PII).

Approvals for Human Experimentation

Before collecting data:

- Obtain IRB approval by detailing the study purpose, risks, and data usage in the consent form.
- MIT COUHES reviews experiment to ensure subjects' rights and privacy are protected.

Consent Form Essentials:

- Purpose: Clearly state the study's intent.
- Risks: Outline potential risks and mitigation.
- Data usage: Specify how data will be used, including any commercial exploitation.

Transformer Stage IV — Tinkering: AI Cancers

- Software methodology: Stage IV uses React for apps, TensorFlow & GitHub for deep learning, Azure & NVIDIA GPUs for computations.
- Testing and experimental plan: Focuses on academic evaluation of OVBM architectures, removing biomarkers, and transfer learning to optimize performance.
- Experiment insights: Base case has a 65% discrimination rate; transfer learning and biomarkers like cough improve results.
- AI concerns: Addressed with explainability tools and bias data; unexpected, better performance in asymptomatic COVID patients. Augmentation of dataset using Biomarker

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Implications of Using MTurk for Crowdsourcing Data

Pros:

- Cost-effective: Cheaper than full-time hires or traditional research.
- Scalable: Easily expand tasks to many workers quickly.
- Diverse workforce: Access to varied skills and perspectives.

- Flexible: Handles a range of tasks from data validation to content moderation.

Cons:

- Quality control: Variable work quality due to diverse skill levels.
- Lack of engagement: Workers may be less motivated, impacting data quality.
- Ethical concerns: Low pay and job insecurity for workers.
- Limited complexity: Not ideal for highly specialized tasks.

